

Lab 03 Exploring Neural Information Processing

Written Analysis

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In this lab, I was able to implement both the biological neuron model and the artificial neuron model. Through the simulation, it was possible to observe the major differences between the biological and artificial systems in how they process information.

In the biological neuron model, the results were clear in how the action potential works. For the low currents (0.2), the membrane potential was slightly higher, though it was not enough to be considered an action potential, which is -55. However, when the currents were higher (0.8), the neuron was able to generate an action potential, which was around 40. At higher currents (1.5), the neuron was able to generate more spikes. This is clear in how the biological neuron works. The stronger the currents, the more spikes the neuron generates.

The graph also illustrated the "all or none" principle. Once the threshold was met, a full spike was generated instead of a partial spike. The membrane potential also reset itself towards the resting potential of -70 mV due to the refractory period after each spike. This made the simulation more realistic.

Unlike the biological neuron, the artificial neuron does not have spikes. Instead, a weighted sum of the input is calculated and an activation function is applied. The comparison of the activation function illustrated that sigmoid has a smooth curve between 0 and 1, whereas ReLU has a linear increase from negative infinity towards positive infinity, and tanh has a range of -1 and 1 and is

symmetric about zero. Unlike biological neurons, artificial neurons have continuous values instead of spikes.

For the artificial feedforward network, different combinations of input resulted in different numerical values of continuous output. The network processed all this information in an instantaneous manner, and there was no concept of time, spikes, or a refractory period.

In the biological network simulation, Neuron 1 continued to spike, but Neuron 2 did not spike since the signal that it was receiving was of insufficient strength. This shows that biological networks rely on signal strength and threshold accumulation. If a signal is insufficient, a neuron will not spike.

In summary, this lab experiment has demonstrated that artificial neural networks are modeled on biological neurons but are much simplified mathematical constructs. Biological neurons function in time and communicate through spikes, whereas artificial neurons function through continuous mathematical functions. Knowledge of biological neural function can help inspire the development of more sophisticated AI models, such as spiking neural networks, that take into account timing and energy efficiency like the biological brain.